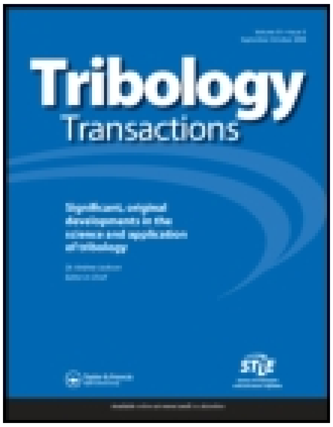

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A Mathematical Model of Spalling Fatigue Failure in Rolling Contact

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A Mathematical Model of Spalling Fatigue Failure in Rolling Contact

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Variables affecting the fatigue life of a rolling contact are identified. A mathematical model of subsurface and surface crack propagation is presented. The life to failure of volume elements in the vicinity of a defect (defect life) is formulated. A term "severity" of a microdefect has been defined. The model is characterized by the inclusion of bulk material parameters, defect characteristics and parameters of geometry, stress, lubrication and surface topography. A statistical expression for the life of an entire rolling body is based on the defect life formula. The new model comprises current standard bearing life prediction formulas as a special case.

Nomenclature

A = Instantaneous crack size
A_c = Critical crack size at end of Phase II
A_p = Self-propagating crack size at end of Phase I
c = Constant exponent
D = Ductility
d = Defect severity (original)
E = Reduced Young's modulus
e = Weibull dispersion parameter
$F(d)$ = Defect severity distribution
f = Function sign
$G(N)$ = Cumulative probability of failure at a defect within N cycles
h = Constant exponent
k_1 = Constant
$H(N)$ = Cumulative probability of rolling body failure within N cycles
M = Material parameters
m = Number of cells
N = Number of stress cycles
N_L = Number of cycles to failure
N_0 = Minimum life
N^* = Characteristic life (scale parameter of the life distribution)
S = Size of highly stressed area in the cross section of a rolling element
$S(N)$ = Cumulative probability of rolling body survival through N cycles
V = Stressed volume
W = Load
X_i = Undefined parameters
$\mathbf{x}$ = Position vector
γ = Matrix strength function
$\Gamma(d)$ = Function relating Phase I crack growth rate to defect severity
Λ = Function sign
ϵ_c = Plastic strain at defect
ϵ_0 = Total microstrain
Θ = Defect severity (as a function of cycling)
φ = Function sign
σ_y = (Micro) Yield stress
σ_n = Hydrostatic compressive stress
ρ = Contact geometry parameters
τ_0 = Amplitude of maximum reversing shear stress in Hertzian contact
z_0 = Depth from rolling surface, at which τ_0 occurs
Π = Multiplication operator

SUBSCRIPTS

I, II, III = Pertaining to Phase I, II, and III of crack growth respectively
i = Pertaining to ith cell
v = Pertaining to volume element
s = Pertaining to surface element

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Introduction

ROLLING bearings are highly engineered devices, and great importance attaches to their reliability. Because of the variety and severity of their operating conditions, such bearings must be expected to have finite (if generally long) life, which is terminated eventually by one of several failure modes recently reviewed in (*1*). Rolling bearing service life is a statistical phenomenon; of a number of apparently identical bearings operated under the same conditions, each can be expected to have a different life-span. Thus, a probability of survival (reliability) attaches to any given design life for a set of bearings and operating conditions.

Among the failure modes discussed in (*1*), most are "preventable" in the engineering sense, i.e., reliability essentially equal to unity can be achieved for practical design lives by suitable design, selection of operating conditions and attention to quality. While this does not assure total absence of failures from these causes, it does focus engineering attention on prevention. There remains, however, one failure mode, viz., spalling fatigue, which is *not*, as far as is known, totally preventable for realistic loads and lives, and attention thus focuses on reliability increase at given design life or longer life with given reliability.

As is well known, rolling bearing spalling fatigue life prediction is the ASA standard method of rating (design selection) for rolling bearings, based on the (tacit) assumption that all other failure modes have been "designed out" of the bearing application so that its reliability is determined by spalling fatigue life. The basis of the standard, and the most complete published method of rolling bearing (fatigue) life prediction is that by Lundberg and Palmgren (*2*, *3*) which has been statistically evaluated for standardization purposes by Lieblein and Zelen (*4*). While several papers have dealt with particular aspects of the Lundberg-Palmgren theory (*5–7*), and the number of application papers is very large, there is, in the published literature, no fundamental reevaluation of the Lundberg-Palmgren premises. Since two decades have lapsed since (*2*) was published and great strides have been made in the study of rolling contact fatigue phenomena (as reviewed e.g., in (*1*)) and in the field of metal fatigue in general, such a reevaluation is clearly justified.

A study was undertaken by the authors with the aim of such a reevaluation. The first year of effort, covering conceptual modeling only, has been reported in (*8*) and is summarized in this paper.

The authors do not challenge or aim to supplant the current ASA bearing rating standard which is experimentally well supported. What is intended is an open-minded reexamination of the physical occurrences in rolling contact fatigue and, hopefully, the creation of broader, more general expressions, permitting account to be taken of many of the variables influencing failure, some of which have only recently been recognized. The Lundberg-Palmgren formulas appear as a special case of the new expressions.

It may be well to state the rationale for the publication of a *conceptual* model of rolling contact fatigue life, without awaiting the development of the specific functional forms and numerical constants of the model. Clearly, without these specifics, the model cannot be verified quantitatively by experiment and is not ready for engineering use. However, the fatigue failure of metals is complex and incompletely understood in most of its aspects and certainly in rolling contact. Therefore, any mathematical model of rolling contact fatigue is subject to controversy as to its fundamental assumptions and its intrinsic logic. The model proposed herein is presented in a form which should permit, by comparison with successful models of other types of fatigue by direct analysis of its logic, a detailed critique on both of these counts.

The principle of rolling bearing life prediction

Rolling bearing fatigue life is predicted on the basis of a cumulative damage concept. As formulated by Lundberg and Palmgren, this concept revolves around the following equation:

$$\log \frac{1}{S(N)} \sim N^e \varphi^e(\tau_0, z_0) V \qquad [1]$$

Physically, Eq. [1] teaches that the cumulative probability of survival decreases with increasing number of cycles N, and with increasing size of the rolling contact system (stressed volume V). The specific choice of the function $\varphi(\tau_0, z_0)$ was made by Lundberg and Palmgren as follows:

$$\varphi^e(\tau_0, z_0) = \tau_0^c z_0^{-h} \qquad [2]$$
$$c,\ e,\ h\text{: constants} > 0$$

Equation [1] is readily modified to take account of time-variable or space-variable loading by using the "Palmgren-Miner hypothesis" of damage accumulation (*9*, *10*), stating that fatigue damage accumulates at a rate depending only on load conditions at the current time, i.e., Eq. [1] is generalized to:

$$\log \frac{1}{S(N)} \sim \int_{(V)} \left[\int_{(N)} \varphi(\tau_0, z_0)\, dN \right]^e dV. \qquad [3]$$

It is noted that Eq. [1] is equivalent to:

$$H(N) = 1 - S(N) = 1 - \exp\left\{ - \left(\frac{N}{N^*} \right)^e \right\} \qquad [4]$$

with

$$N^* \sim \tau_0^{-c/e} z_0^{h/e} V^{-1/e},$$

i.e., lives to failure follow a Weibull distribution with zero lower bound, characteristic life N^* and dispersion exponent e.

The phenomenological nature of Eqs. [1] through [4] results in an impasse if one attempts to incorporate into life prediction newly acquired knowledge regarding the effect of parameters other than contact geometry and kinematics, since these equations offer no clue as to the proper role of such parameters in defining life.

Therefore, in the present study, novel equations are being generated from which the Lundberg-Palmgren equations can be obtained as a special case.

Principal variables of fatigue failure

Chiu, Martin, and McCool (*8*) give a review of the variables governing a rolling contact fatigue failure situation. They fall into four main categories: material variables, surface microgeometry variables, design variables and operating variables.

Material variables are those influencing the "strength" of the rolling system. Current fatigue investigations (chiefly of the non-rolling type) consider yield strength and ductility of the material as dominant bulk (or matrix) strength variables (*11–13*). Modifying these are residual stresses and work hardening effects acquired, in part, during the course of fatigue life. In rolling contact, the materials used are of high hardness, and such materials do not react with their matrix strength. Rather, rolling contact life appears to be determined by the strength of the material in the vicinity of inevitable material imperfections such as inclusions in the matrix or microcracks (*14, 15*). Thus, the nature of these imperfections is a dominant variable.

Surface microgeometry variables determine the detailed nature of the contact through which loads are transmitted to the material. The generalized roughness of the surface determines the topography of the contacting surface elements as well as the plastic behavior of the material immediately adjacent to the surface, and interacts with lubrication (*16*).

There are localized imperfections on surfaces, mostly in the form of sharp depressions ("furrows") which form stress raisers near their edges and are influential in failure (*15, 17*). Of course, there can be many other types of surface variables, some of them artificial as induced by coatings, treatments of the surface, etc.

The *design variables* of rolling contact, dealt with by the Lundberg-Palmgren theory, are familiar. Track ength, dimensions and number of rolling elements, lcontact angles, conformity parameters (defining the precise cross track geometry) and many others are influential, primarily because they determine the macroscopic (Hertzian) stress field and the number of cycles as a function of bearing ring rotation. They also determine the magnitude of the highly stressed volume.

Operating variables encompass the external environment in which the rolling contact system must endure, including load, speed, lubrication, temperature, and atmospheric conditions. The influence of load in determining the stress field is obvious, and so is that of speed in determining the number of cycles per unit time. However, these two variables also interact with the lubricant to determine the hydrodynamics of the often-present pressure-bearing elastohydrodynamic (EHD) lubricant film in the contact, which redistributes stresses and has important effects on the microbehavior of the contact area (asperity interaction) and on life (*18–20*). Temperature enters by influencing both the material strength and the lubrication, and atmospheric conditions can be of consequence if they influence lubricant behavior or cause corrosive effects.

The fatigue failure model

The proposed failure model visualizes fatigue damage as the growth of a crack as a result of cyclic stressing. There are dislocation motions, plastic flow occurrences, and carbon migration in the matrix which precede or are concurrent with crack formation. However, these subtler occurrences are not helpful for modeling purposes because no way is known to measure the degree to which they are shortening life expectancy. The effect of a crack on life is intuitively clear: when the crack has become large enough, a piece of material will separate from the surface, forming the fatigue spall defined as rolling contact fatigue failure. Crack growth is visualized as irreversible progress towards failure. The fatigue phenomenon starts, accordingly, with the initiation of a crack, and proceeds through stages of its growth until the crack has become large enough to form a spall. Fatigue life, as determined by the crack in question, begins with the onset of cycling and terminates when the spall forms. Of course, a rolling contact system may be functional in the presence of a spall of tolerable size. There is room to accommodate this argument in the model, but the current description assumes that the failure has occurred with first spalling.

It is convenient to describe failure generation in three phases:

Phase I begins immediately upon the onset of cyclic stressing, and is consumed by the formation and growth of a microcrack to the point where the crack becomes self propagating as defined below.

In *Phase II*, the crack grows macroscopically until it has reached a critical size large enough to initiate "precipitous" crack growth.

Phase III is occupied by precipitous crack growth at a rate greatly in excess of Phase II growth. The precipitous cracking forms the spall. This Phase may be virtually instantaneous or consume substantial length of time, depending on whether one specifies a minimum spall size which is accepted as a failure, and depending on a variety of material and operating conditions.

Mathematical Life Formula

A mathematical life formula describes crack growth through the above three Phases. In order to do so, one selects a measure of crack size A and describes it as a function of the number of stress cycles N in the form

$$A = f_1(N, X_i) \qquad [5]$$

The X_i are as yet undefined parameters.

At the present state of knowledge, it seems best to select a form of Eq. [5] common in current fatigue theory (*12*) defining the crack growth rate dA/dN in terms of the relevant variables:

$$\frac{dA}{dN} = f_2(N, A, X_i). \qquad [6]$$

Life is predicted by determining from Eq. [6] that value of the number of cycles N_L which corresponds to a critical crack size A_c, causing immediate spalling, i.e.,

$$N_L = f_3(A_c, X_i). \qquad [7]$$

A group of theories of fatigue describe the crack growth rate as a function of the following simpler form:

$$\frac{dA}{dN} = \Lambda(\epsilon_c, D) \equiv \Lambda[\epsilon_c(N), D(N)] \qquad [8]$$

The only variables entering the crack propagation Eq. [8] are a *plastic* strain ϵ_c (measured at the propagating crack front) and a measure D of material ductility.

Note that Eq. [8], for all its simplicity, contains many restrictions. Only the first derivative of crack size appears explicitly. (The crack size itself enters by way of its influence on ϵ_c.) Only variables measurable at the location of the propagating crack front appear, and these only with their values assumed at the time of the Nth stress cycle. (Of course, there is a dependence of $\epsilon_c(N)$ and $D(N)$, and the specific form of this dependence determines whether this formula satisfies the Palmgren-Miner hypothesis.)

In order to add definition to Eq. [8], the concept of a "defect" is introduced. A defect exists at a point in the material or at the surface. At that point, there is a tendency of crack generation. This tendency manifests itself in Eq. [8] by causing ϵ_c to be higher at the defect than elsewhere in the vicinity.

An expression for ϵ_c at a defect can be written by *assuming* that one can select (as in uniaxial tension) a critical scalar ϵ_0 of the *total* (elastic plus plastic) macroscopic strain field as it would exist at the defect location, in the absence of the defect, such that ϵ_c depends only on ϵ_0, on defect "severity" θ and on the yield strength σ_y of the matrix, i.e.,[1]

$$\epsilon_c = f(\Theta,\epsilon_0,\sigma_y) \qquad [9]$$

The defect severity factor Θ is defined as the "strain raising" factor of the defect. An "ineffective defect," i.e., one which does not raise the magnitude of the strain, is assigned $\theta = 1$. The severity Θ is a function of N.

Introducing Eq. [9] into Eq. [8], one has

$$\frac{dA}{dN} = \Lambda(\Theta,\epsilon_0,\sigma_y,D) \qquad [10]$$

In Eq. [10], Θ is the one variable related to defects, whereas ϵ_0, σ_y and D are related to the matrix. For simplicity, the matrix effects can be consolidated into a single function γ, i.e.,

$$\gamma = \gamma(\epsilon_0,\sigma_y,D) \qquad [11]$$

The new assumptions underlying Eqs. [9] through [11] are that there is a scalar ϵ_0 of the macrostrain field which, alone among strain field characteristics, determines ϵ_c and that all matrix parameters exert their influence on crack growth via a single quantity γ.

Further development of Eq. [11] requires use of a further restrictive concept such as the hypothesis of multiplicative effects on fatigue life. This concept, also used by Lundberg and Palmgren, asserts that the rate of fatigue damage (crack growth) can be expressed as a product of a number of *independent* factors φ_i, i.e.,

$$\frac{dA}{dN} = \prod_i (\varphi_i^{a_i}),\ a_i = \text{const.} \qquad [12]$$

[1] Generally, of course, ϵ_c depends on the three principal macrostrains $(\epsilon_0)_i$; $i = 1, 2, 3$ and Θ may be orientation dependent, i.e., there exists Θ_i; $i = 1, 2, 3$.

Applying this concept to Eq. [11], one may, by suitable definition, absorb the function Λ in the functions Θ and γ and write:

$$\frac{dA}{dN} = \Theta \cdot \gamma \qquad [13]$$

The matrix factor γ

γ is defined in Eq. [11] as a function of a critical total macrostrain scalar at the location of the growing crack front, and of a yield strength and a ductility measure.

The yield strength measure σ_y is a microyield stress (*21*) since small scale plastic occurrences are at issue in rolling contact fatigue. It must be taken with its value at the time of the Nth stress cycle, to account for work hardening or work softening of the matrix.

It is intuitively convincing that one should include in the life formula a material ductility property measuring the amount of plastic strain the matrix can absorb before it cracks. A variety of metallurgical parameters, but also some operating conditions, will determine ductility. The most important operating condition is hydrostatic compressive stress σ_h (*22*). It is generally believed that the high hydrostatic compression component existing over most of the Hertzian contact stress field retards crack formation (*7*). This fact will manifest itself in a point-wise varying value of the ductility parameter D when examining material elements located at different points within the Hertzian stress field. Inasmuch as it may depend on work hardening, ductility can be a function of N. Thus, the ductility parameter is known to depend on material factors (symbolized by M) on a parameter of the stress field σ_h, and can depend on N. It may also be related to other operational parameters (e.g., to cycling rate) generally designated φ. These relationships are symbolized by the following equation:

$$D = D(M,\sigma_h,N,\varphi). \qquad [14]$$

Turning to the critical macrostrain parameter ϵ_0, it is obviously dependent on the variables of load W, contact geometry ρ, position under the contact $\mathbf{x}$, and elastic modulus E, defining between them the elastic Hertz stress field. With a quasi-elastic assumption, these parameters give a relationship of the form

$$\epsilon_0 = \epsilon_0(W,E,\mathbf{x},\rho). \qquad [15]$$

The "quasi-elastic" assumption operates on the scale of the whole Hertzian stress field and disregards the vicinity of defects. It postulates that the macroscopic total strain ϵ_0 can be calculated from the elastic (Hertzian) stress field of the rolling contact. This is the case for all practical bearing loads, since they are sufficiently low that, at most, very small amounts of macroscopic plastic flow take place (*23*), so that the plastic component of total strain is negligible, and that plastic flow does not result in a significant redistribution of elastic stresses. Except for the generation of residual stresses due to cyclic stressing, this is a reasonable assumption in all practical rolling contact fatigue situations.

The question of residual stresses may be handled by assuming that, after a small number of cycles, they have "shaken down" to a constant value. Then, they act as a superimposed static stress field and combine with the cyclic stresses. The resulting time-variable stress field is different from that existing without residual stresses, and assumptions must be made regarding the effect of this difference on crack propagation.

Quasi-elastic behavior is restricted to the matrix at locations remote from defects. Microplasticity *at defects* is the condition of cracking in the proposed model. For any given defect (Θ) and given matrix strength (σ_y), it is possible to delineate that portion of the Hertzian stress field within which the macrostrain ϵ_0 is high enough to cause plastic microstrain ϵ_c. For a given population of defects, there will be a "realistic" maximum severity Θ. One can delineate a highly stressed area in the Hertz stress field within which all microplastic occurrences occasioned by defects of "realistic" severity will be confined. This defines a "highly stressed zone" and the cross sectional area of this highly stressed zone, in a plane perpendicular to the rolling direction, will be designated by S.

The defect severity factor Θ

The effect of a "defect" in generating plastic strain in its vicinity is manifestly complex. A relationship exists between Θ and N because growth of a crack redistributes the stresses around the defect. This can be expressed as follows if the initial defect severity is d:

$$\Theta(N) = \Theta(d,A(N),S). \quad [16]$$

As mentioned before, crack propagation can be envisioned in three phases. In the first phase, microcrack propagation, the crack is so small that, by comparison, the size of the highly stressed zone can be considered infinite, so that for Phase I, Eq. [16] is simplified to

$$\Theta_{\mathrm{I}} = \Theta_{\mathrm{I}}[d,A(N)]; A \leq A_p. \quad [17]$$

Here, A_p is the self-propagating crack size at the end of Phase I (see below).

Phase II starts at that point in time when the crack has grown sufficiently large to outweigh, in its effect on propagation rate, the original defect. Such a crack does not require the defect to propagate; it is "self-propagating." For a crack of this size, or larger, the size of the highly stressed zone can no longer be considered infinite, so that one has:

$$\Theta_{\mathrm{II}} = \Theta_{\mathrm{II}}(A(N),S); A_p \leq A \leq A_c. \quad [18]$$

A crack of size A_c, defining the end of Phase II, is assumed to lead to a spall "instantaneously" by a precipitous fracture mechanism. The rolling contact system does not necessarily become inoperative immediately when this crack size is reached. However, there is a "catastrophic" growth step between the crack of size A_c and the completed spall, i.e.,

$$\frac{dA}{dN} \to \infty \text{ for } A \to A_c. \quad [19]$$

One can assume that the critical crack size is related to the size of the highly stressed zone. In the simplest form:

$$A_c = k_1 S, k_1 = \text{const.} \quad [20]$$

Prediction of the life at a defect

Substituting Eqs. [17] and [18], respectively, into Eq. [13], one obtains the following formulas for crack propagation rate during Phases I and II:

$$\left(\frac{dA}{dN}\right)_{\mathrm{I}} = \gamma_{\mathrm{I}}(\epsilon_0,\sigma_y,D) \cdot \Theta_{\mathrm{I}}[d,A(N)]; A \leq A_p \quad [21a]$$

$$\left(\frac{dA}{dN}\right)_{\mathrm{II}} = \gamma_{\mathrm{II}}(\epsilon_0,\sigma_y,D) \cdot \Theta_{\mathrm{II}}(A(N),S); A_p \leq A \leq A_c, \quad [21b]$$

where subscripts I and II apply to Phases I and II respectively.

Using the previously explained multiplicative hypothesis on the function Θ, these equations may be rewritten as follows:

$$\left(\frac{dA}{dN}\right)_{\mathrm{I}} = \gamma_{\mathrm{I}}(\epsilon_0,\sigma_y,D) \cdot f_1(A) \cdot \Gamma(d); A \leq A_p \quad [22a]$$

$$\left(\frac{dA}{dN}\right)_{\mathrm{II}} = \gamma_{\mathrm{II}}(\epsilon_0,\sigma_y,D) \cdot f_2(A/S) \cdot f_3(A); A_p \leq A \leq A_c \quad [22b]$$

Γ is an (increasing) function of initial defect severity *d only*.

In Eq. [22b], two functions f_2 and f_3 are shown, one representing the effect of relative crack size by comparison to the size of the highly stressed zone, and the other any remaining direct effect of absolute crack size (as hypothesized e.g., by Lundberg and Palmgren when introducing the effect of the depth co-ordinate z_0 of the maximum alternating shear stress amplitude). Note that in Eq. [22b] the original defect severity d does not appear. Therefore, this equation contains only macrostrain and matrix variables, and is independent of the original defect population. In Eq. [22a] on the other hand, $\Gamma(d)$ has a different value for each individual defect, and the equation is, therefore, dependent on the defect population.

Integration of the differential equation [22a] and substitution of A_p yields a value N_{I}, the life at the end of Phase I. Integration and substitution of A_c yields, from Eq. [23b] a value N_{II}, the duration of Phase II life. The life N_L from the beginning of cycling through the end of Phase II is then obtained as the sum of these two phase-lives $N_L = N_{\mathrm{I}} + N_{\mathrm{II}}$. (Phase-III life may or may not be 0, depending on the definition of failure.)

The expressions for N_{I}, N_{II}, and N_L are:

$$N_{\mathrm{I}} = \frac{f_{\mathrm{I}}(A_p)}{\gamma_{\mathrm{I}} \cdot \Gamma(d)} \tag{23a}$$

$$N_{\mathrm{II}} = \frac{f_{\mathrm{II}}(A_c, A_c/S)}{\gamma_{\mathrm{II}}} \tag{23b}$$

$$N_L = N_{\mathrm{I}} + N_{\mathrm{II}}. \tag{23c}$$

The apparent arbitrariness in the selection of the self-propagating crack size A_p will not influence the total life $N_L = N_{\mathrm{I}} + N_{\mathrm{II}}$ if our hypotheses are correct, because Eqs. [22a] and [22b] were both obtained from Eq. [16] by neglecting, for Eq. [22a], the influence of S and for Eq. [22b] the influence of d. Inasmuch as these approximations are valid, the two equations merely describe two portions of the same function $A(N)$, and their domains of validity overlap so that the selection of A_p is, within limits, discretionary.

After having selected a suitable value of A_p and determined that value of A_c/S at which the crack becomes critical, only γ_1 and d remain in Eq. [23a] as functions of external parameters and S and γ_2 remain in Eq. [23b]. All these remaining parameters are observable.

The statistics of life of an entire rolling body

In determining the life of a finite-size rolling body, one starts with Eqs. [23a–c] for life in the vicinity of a given defect and uses statistical theory to obtain life for a volume of material containing a multitude of defects.

Failure of the rolling body will occur through "competition" between a multitude of potential defects acting as failure nuclei. In Phase I, microcracks grow at a multitude of defects, at differing rates, independently of each other. One of these microcracks, or several, will reach the beginning of Phase II within the life of the part. These proceed to accumulate Phase-II life until one of them has generated a crack of critical size A_p, at which a spall forms, whereupon the rolling body is considered failed. All other defects which have entered Phase II show, at the time of failure, cracks of less than critical size. In the proposed model, variations in original matrix strength within a rolling body are neglected. This leaves two main sources of variability: the systematic point-wise variability of the macrostrain field in the contact zone (and the consequent variability of work hardening and residual stresses) and a random variability of defect severity and location within the contact zone. The effect of these variables on the life calculated from Eqs. [23a] and [23b] determine the outcome of the competition among defects for the generation of the crack leading to failure.

A statistical treatment describing this competition will be illustrated for Phase I. Phase-II life will be considered a deterministic quantity, calculable for each point in the rolling body, from the knowledge of macroscopic strain and matrix parameters alone.

To express the statistics of Phase-I life, consider the highly stressed zone to be composed of elementary "cells" of uniform size, small enough to contain at most one effective defect with $d > 1$, but large enough for a crack of size A_p to be wholly confined within the cell. Then, Phase I fatigue damage, originated within a cell, will remain confined within it and will not influence the behavior of adjacent cells. On this assumption, Phase-I life of each cell is independent of the life of all other cells.

Recalling that "defects" with severity $\Theta = 1$ (no strain raising properties) have been defined, one can now say that each cell contains at the start of cycling *exactly* one "defect" with severity $d \geqq 1$. Since only defects with $d > 1$ generate a crack, the cell will, during cycling, contain *exactly* one "defect" with severity $\Theta \geqq 1$. Assume now that there is a known cumulative distribution $F(d)$ of original defect severities d for each cell:

$$\mathrm{Prob}(d_1 \leq d) = F(d), d \geq 1. \tag{24}$$

Given $F(d)$, Eq. [23a] as transformation equation between N and d, permits determination of a probability distribution $G(N)$ for Phase-I life N_{I} of an individual cell:

$$\mathrm{Prob}(N_1 \leq N_{\mathrm{I}}) = G(N_{\mathrm{I}}|\mathbf{x}) \tag{25}$$

In the general case the functions f_{I}, γ_1 and Γ in Eq. [23] vary from point to point in the rolling body because of the nonuniformity of the macrostrain field (or for other reasons), and the distribution $G(N_{\mathrm{I}})$ will depend on the position vector $\mathbf{x}$.

From Eq. [25] statements can be derived regarding the probability of failure of the entire rolling body. The rolling body will fail if exactly one of its cells fails. Therefore, the probability distribution is needed for the life of that cell which fails first of all cells in the rolling body.

If the fatigue phenomena in each cell are independent, the probability that the rolling body *survives* is the product of the survival probabilities of all cells in it, i.e., the (Phase I) life distribution $H(N_{\mathrm{I}})$ of an entire rolling body is:

$$H(N_{\mathrm{I}}) = 1 - \prod_i [1 - G_i(N_{\mathrm{I}})]. \tag{26}$$

Equation [26] follows from Eq. [25] because the probability of survival is obtained by subtracting the probability of failure from unity.

A special case of Eq. [26] arises if all G_i are identical. This is the case for a thrust loaded bearing ring (the Hertz stress field is independent of contact position) in which there is only one type of defect distribution throughout the ring and the defects are infrequent so that a "cell" can be represented by a short "slice" of ring between two closely spaced cross-sections. If there are m such slices, Eq. [26] reduces to

$$H(N_{\rm I}) = 1 - [1 - G(N_{\rm I})]^m. \quad [27]$$

As shown in (8), Eq. [27] asymptotically approaches, for large m, the form of a Weibull distribution with dispersion exponent e:

$$H(N_{\rm I}) = 1 - \exp\left(-\left\{\frac{N_{\rm I} - N_0}{N^*}\right\}^e\right); N^* \sim m^{-1/e} \quad [28]$$

provided that $G(N_{\rm I})$ falls within a general class of functions for which $G(N_{\rm I}) \sim N_{\rm I}{}^e$ if $N_{\rm I} \to N_0$.

The constants are determined by the specifics of the distribution function $G(N_{\rm I})$ which, of course, depends on cell size and thus, for a given rolling body, varies with m. N_0, the minimum Phase-I life, can be considered zero, since a crack of size A_p can pre-exist in the matrix. Eq. [28] emerges from general theorems on the asymptotic properties of extreme value distributions and is not a separate hypothesis (8).

The (Phase I) life distribution of a rolling body as given in Eq. [28] is not merely observed as a phenomenological fact, it is related to the physically meaningful distribution $F(d)$ of defects. The relationship between the distributions $F(d)$ in Eq. [26] and $G(N_{\rm I})$ in Eq. [25] permits inferences from one distribution to the other, such as the identification of appropriate measures of defect severity. It is also possible to conjecture appropriate defect severity distributions and test these conjectures by the effect they have on the life distribution of the part (8).

Equation [28] contains a volume effect on Phase-I life. For cells of fixed size, their number is proportional to the stressed volume. Therefore

$$N^* \sim V^{-1/e}, \quad [29]$$

i.e., the scale parameter of the Weibull life distribution is an inverse function of the stressed volume as in the Lundberg-Palmgren theory.

The observable life N_L of an entire rolling body is, according to Eq. [23c], the sum of Phase-I and Phase-II lives at the defect where N_L *will be shortest*. This can be calculated *simply*, only under one of the following two alternate conditions for Phase-II life $N_{\rm II}$:

(a) $N_{\rm II} \ll N_{\rm I}$, or
(b) $N_{\rm II} \approx$ constant.

For either condition, the shortest N_L will coincide with the shortest $N_{\rm I}$, so that $N_{\rm I}$ can be taken from Eqs. [26], [27], or [28]. $N_{\rm II}$ is then calculated from Eq. [23b] for the $\mathbf{x}$ at which the shortest $N_{\rm I}$ has occurred and the two are added.

If neither of the above conditions for $N_{\rm II}$ is satisfied, then the (deterministic) variation of $N_{\rm II}$ with $\mathbf{x}$ may cause the shortest N_L to occur at an $\mathbf{x}$ different from that at which occurred the shortest $N_{\rm I}$. For this case, the statistical treatment must include $N_{\rm II}$, and this may lead to difficulties in satisfying the assumption of independence of all failures required for Eq. [26].

Application of the life prediction equations

Equations [23a, 23b, and 23c] and the statistical interpretation contained in Eqs. [24–28], represent the proposed fatigue life prediction model. The approaches that can be taken towards application of the model will now be illustrated by a few examples.

THE LUNDBERG-PALMGREN CASE

The following is an example of *one* of several possible methods by which the Lundberg-Palmgren formulas can be obtained as a special case of the proposed model.

Assume in Eqs. [23b] and [23c] that $N_{\rm II} \approx 0$. In Eq. [23a] assume that A_p and $\Gamma(d)$ are defined by the material used; and that, because in Eq. [11] σ_y and D are material constants, $\gamma_{\rm I}$ depends only on ϵ_0, which in turn is, for the entire highly stressed volume, determined by the maximum alternating shear stress amplitude τ_0. Then one obtains

$$N_L \sim \frac{f_{\rm I}\ (\text{const.})}{\gamma(\tau_0)\Gamma(d)} \text{ or } N_L\gamma(\tau_0)\Gamma(d) = \text{const.} \quad [30]$$

If one uses a power function to express γ in terms of τ_0, then

$$N_L^e\tau_0^c\Gamma(d) = \text{const.} \quad [31]$$

The designation of the constant exponents is that used in Eqs. [1] and [2].

According to Eqs. [1] and [2], the depth z_0 at which the stress τ_0 arises would enter the life formula, whereas in Eq. [31] it does not. This contradiction is not surprising in view of the discussions given with Eqs. [23a–23c]: there it was assumed that the Phase-II life depends on A_c/S but that Phase-I life does not. This is justified only if $A_p \ll A_c$ which is contrary to the hypothesis $N_{\rm II} \approx 0$ just made, because crack growth in Phase II cannot, in essentially zero time, cover a large size jump from A_p to A_c in a *fatigue* propagation mode. Thus, if it is desired to maintain that $N_{\rm II} \approx 0$, it is necessary to assume $A_p \approx A_c$ and Eq. [16] should be substituted into Eq. [13] to express $N_{\rm I}$ instead of substituting Eq. [17] into it as was done before. Then, Eq. [23a] will be replaced for the Lundberg-Palmgren case by:

$$N_{\rm I} = \frac{f_{\rm I}(A_p,S)}{\gamma_{\rm I}\cdot\Gamma(d)} \quad [32]$$

Assuming that S is measured by z_0, Eq. [30] is replaced by

$$N_L\gamma(\tau_0)f_{\rm I}(z_0)\Gamma(d) = \text{const.} \quad [33]$$

or using power functions

$$N_L^e\frac{\tau_0^c}{z_0^h}\Gamma(d) = \text{const.} \quad [34]$$

Assuming with Lundberg and Palmgren that the defect severity distribution $\Gamma(d)$ of Eq. [34] is identical for all samples of the contact material, Eq. [34] used as transformation equation will produce a defect life distribution $G(N_L)$ in which

$$\left(\frac{\tau_0^{c/e}}{z_0^{h/e}}\right)^{-1}$$

is a scale factor. Due to Eq. [26], the same scale factor will appear in the rolling body life distribution $H(N_L)$. Equation [29] continues to hold so that $V^{-1/e}$ is a volume-dependent scale factor. Thus

$$N^* \sim \left(\frac{\tau_0^{c/e}}{z_0^{h/e}}\right)^{-1} V^{-1/e} \qquad [35]$$

yielding the Lundberg-Palmgren Eq. [4].

Deviations from the Lundberg-Palmgren Type Weibull Distribution with $N_0 = 0$

It has been found experimentally (*9*) that in Eq. [28], $N_0 > 0$, i.e., there is a non-zero minimum bearing life prior to which there is no finite probability of fatigue failure. This is not explicable by the Lundberg-Palmgren theory. However, it is immediately obtained if Eq. [23b] is assumed to give a non-zero value for N_{II}. In this case, N_I may still be distributed according to a Weibull distribution with $N_0 = 0$, but the total life to failure N_L will have a positive minimum value.

Effect of Material Cleanness (Inclusion Content)

Assume that there are two materials, one "cleaner," the other of lesser cleanness, i.e., possessing different inclusion severity distributions $F(d)$ per Eq. [24], but otherwise identical. The probability of encountering inclusions of great severity is higher for the material of lesser cleanness. One may then expect a larger number of more effective stress raisers in a rolling body of given size, if made of the less clean material. According to Eq. [23a], the relationship between Phase-I life N_I and defect severity is inverse. Consequently, the distribution of life $G(N_I)$ in Eq. [25] will show higher probability for shorter lives in the steel of lower cleanness. The distribution of cell lives will be scaled to lower values. The scale parameter N^* in Eq. [28] will therefore be a smaller number, i.e., the characteristic life of the rolling body will be reduced.

Two Competing Failure Modes

Assume there is not one family of defects (e.g., inclusions) but two families (e.g., inclusions and surface defects). Assume that these two families of defects operate independently of each other and each has a severity distribution. It is possible to define cells such that they have either one or the other type of defect, but not both. This state of affairs is realistic: failure in rolling contact has been shown to initiate either subsurface (e.g., from inclusions) or at the rolling surface (due to surface defects). Cells with inclusions can with little error be considered as volume elements not extending to the surface; cells with surface defects can be considered surface areas underlaid by a thin "skin" of material not overlapping the former cell category. The distribution of the two types of defect is independent.

One obtains two cell life distributions of the type of Eq. [25]: one for inclusions and the other for surface defects:

$$\text{Prob}(N_1 \leq N_{I,V}) = G_V(N_I|x,y,z) \qquad [35a]$$

$$\text{Prob}(N \leq N_{I,S}) = G_S(N_I|x,y). \qquad [35b]$$

Equation [35a] applies to subsurface volumes (three co-ordinates), and Eq. [35b] applies to surface areas (two co-ordinates).

The probability of rolling body failure can be obtained by considering that survival of the rolling body necessitates survival of all cells from both populations. Accordingly, in analogy to Eq. [26]

$$H(N_I) = 1 - \left(\prod_V [1 - G_V(N_I)] \prod_S [1 - G_S(N_I)]\right) \qquad [36]$$

or by analogy with Eq. [27]

$$H(N_I) = 1 - ([1 - G_V(N_I)]^{m_V}[1 - G_S(N_I)]^{m_S}), \qquad [37]$$

where m_V = number of cells with inclusions,
m_S = number of cells with surface defects.

The resulting asymptotic life distribution for a large number of cells is a Weibull distribution with $N_0 = 0$ only if each of the two defect populations yields a Weibull distribution with $N_0 = 0$ and if their dispersion exponents are identical (*24*).

The Effect of Residual Compressive Stresses

If there are residual stresses in the highly stressed zone, their hydrostatic pressure component can be expected to influence ductility. Accordingly, γ_I and/or γ_{II} in Eqs. [23a] and [23b] will change with N. This may have an effect on crack initiation (Phase I) and/or crack propagation (Phase II).

It is also possible to account for the effect of the nonhydrostatic component of residual stresses if a reasonable assumption can be made regarding the effect of a *static* stress component on the microplastic strain ϵ_0. Some fatigue theories offer expressions for this effect (*22*).

Effect of Hardness

Three parameters, ϵ_0, σ_y, and D enter the matrix strength function γ in Eq. [11]. The assumption was made by Lundberg and Palmgren that the only stress-field variables influencing life are the maximum alternating elastic shear stress and its depth co-ordinate. This is tenable only if the material is kept constant so that σ_y and D do not vary. For different materials, γ will depend on the microyield stress σ_y, which is known to be related, although not equivalent, to indentation hardness. Thus, a general expression for γ contains a material strength parameter of the type of hardness, and the form of this expression can be derived using plasticity theory.

Non-Hertzian Contacts

Practical contacts are non-Hertzian in two respects: (*a*) Roller to race contacts are macroscopically non-Hertzian because roller profiles are not correctly approximated by second order surfaces (roller crowning, edge effects); and (*b*) all contacts are microscopically non-Hertzian because of the presence of surface asperities, which will be discussed later.

No Hertzian assumption is inherent to any of the derivations given above. Provided that the total strain ϵ_0 can be defined point-wise in the contact, it is possible to calculate cell life and hence rolling body life, using the proposed model.

Lubrication Effects

The most thoroughly explored lubrication effects to date are those of elastohydrodynamic lubrication, i.e., the transmittal of contact pressure via a pressurized viscous film of lubricant. The effects of such a film on life can be treated on two levels: the macroscopic redistribution of pressure brought about by the elastohydrodynamic film can be used to correct the macroscopic strain field in the matrix which defines the function γ. Microscopically, it is possible to describe the influence of a *partial* elastohydrodynamic film on asperity interactions. (A start has been made in this direction and is described in (*8*).) The result is a modification of the microstress field in the vicinity of the surface, and a new distribution of cell life in the population of cells containing surface defects. In principle, it is thus possible to calculate these elastohydrodynamic effects on rolling body life.

Size Effects

In many fields (small instrument bearings, large rolling mill and radar pedestal bearings) size effects are of great importance. The model presented offers several clues to size effects.

The self-propagating crack size A_p is defined as the smallest crack which progresses at a rate independent of the original defect. The magnitude of A_p must depend on the "radius of effectiveness" of the most severe original defect, i.e., it must be related to the volume within which the stress raising effect of the defect is felt. This volume is likely to be proportional to a defect dimension. Accordingly, A_p depends on the size distribution of the original defects. Since A_p enters into the determination of the Phase-I life of a defect, it produces a size effect on N_{I}. Absolute bearing size will enter into this effect inasmuch as it influences steel processing practices, and thereby the size distribution of defects.

The other Phase-I size effect concerns the cell size introduced in connection with Eq. [24]. Since cells can contain only one defect, their size is related to the defect spacing. This spacing is, again, influenced by absolute bearing size through the correlation between reduction in cross sectional area undergone by the steel during rolling or forging and defect distribution. Cell size indirectly enters Eq. [27] because the number of cells present in a given rolling body depends on their size. As stated before, the distribution $F(d)$ in Eq. [24] also depends on cell size so that the effect is not a simple one.

Given a defect size distribution for a selected cell size, the stressed volume of the rolling body, in terms of multiples of unit cell volume, introduces the volume effect on life, represented by Eq. [29].

Turning now to size effect on Phase-II life, Eq. [23b] shows two size effects. One is represented by the factor A_c. It is the (hypothetical) effect of absolute crack size on propagation rate. The other, more obvious effect, is represented by the ratio A_c/S and represents the fact that a crack must grow to a certain size with reference to the size of the highly stressed zone before a spall can form (the crack must propagate from the depth where it is formed, to the surface, or vice versa, and spalls are typically of a depth comparable to that of the location of one of the maximum shear stresses). This effect is related to absolute bearing size through S.

There are, of course, numerous size effects implied in Eq. [15] relating total macrostrain to external load and to contact geometry. Most of these effects are accounted for in the Lundberg-Palmgren theory. Steel processing effects are also implied in Eq. [15] through the fact that the yield strength σ_y may well depend on the absolute size of the part. The same applies to the ductility parameter D in Eq. [14]. Clearly, the proposed model offers ample room for the exploration of size effects, providing, of course, that the necessary experimental evidence can be secured.

Conclusion

A mathematical model for the prediction of the life of rolling contacts has been proposed, based on a concept of crack propagation from preexisting defects. Numerous parameters characterizing the material and geometry of the contact and the operating conditions are incorporated in the model, including load, lubrication, matrix strength, defect population, and defect severity.

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DISCUSSION

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On the occasion of this re-evaluation and generalization of the Lundberg-Palmgren theory of bearing life prediction it might be suitable to communicate findings on a spalling failure mode which is different from the classical, subsurface originated one. This mode seems to be governed by simple laws, and is also accessible to mathematical analysis. In (*A1*) it was stated that sometimes the prevalent cause of spalling failure is not subsurface inclusions, but a well-known type of rolling bearing raceway imperfection, namely circumferential furrows left behind from the finishing process.

Circumferential furrows on inner ring raceways of 6309 size single row deep groove ball bearings were examined in the discussor's laboratory by microphotographing their environment on the raceway surface and by profile tracing their cross-section before and after endurance runs. In the endurance runs the bearings were operated under the following conditions: load: 4240 lb, radial: speed: 9700 rpm; lubricant: mineral oil of 12 cs viscosity at operating temperature. The corresponding theoretical elasto-hydrodynamic oil film thickness at the inner ring contacts is 20 μin. The typical appearance of a furrow in a fatigue tested bearing is shown in Fig. A1. The furrow is accompanied by two bright bands adjacent to either side of the furrow and parallel to it. The profile tracing shows that these bands are highly burnished ("glazed"): the profiles of the bright bands appear very smooth, compared to the rest of the raceway surface. The tracings also reveal that the glazed bands are inclined planes having a slope towards the furrow of $1.7° \pm 0.2°$ relative to the nominal raceway surface. For exceptionally severe furrows the profiles of the glazed bands have a slight convex curvature. In these cases the average slope is $1.7° \pm 0.2°$. The glazed area and the sloping area coincide. For a particularly large furrow of about 150 μin. depth, a cross-section before and after endurance run is shown in Fig. A2. It was found that during cyclic stressing of the raceway surface the edges of a furrow sink down, while the lower part of the furrow cross-section narrows. In this process the total cross-sectional area

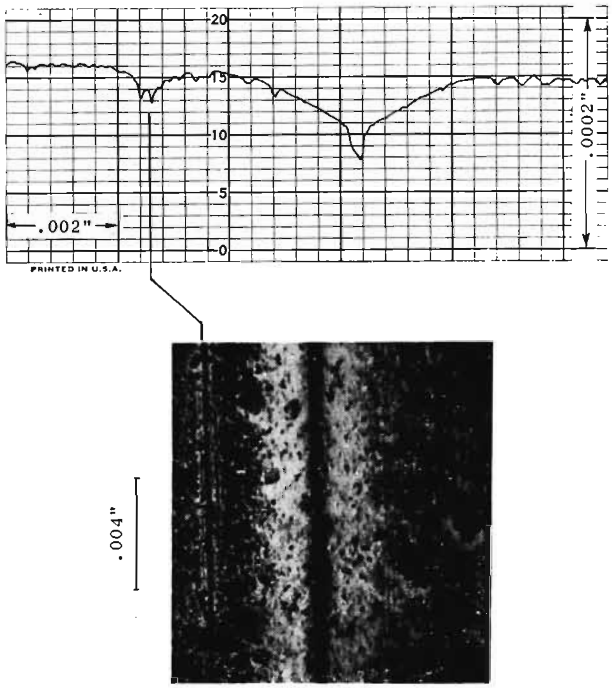

Fig. A1

FIG. A1. Talysurf trace and microphotographs of surface furrow on 6309 inner ring after endurance test. Life: 199 millions rev. Material: 1st remelt consutrode 52100.

of the furrow remains constant within 5%. From this result it is concluded that the sloping glazed bands are formed by plastic deformation. A calculation of the subsurface von Mises shear stress distribution given in (*A2*) for a two-dimensional contact containing a furrow of a size similar to the observed furrow sizes shows that the maximum shear stress is four times as high as that of the same contact without furrow and appears close to the surface beneath the furrow edges. The numerical result depends on the specific profile of the furrow shoulders selected for the calculation. This profile resembles an intermediate condition in the process of plastic deformation of the area near the furrow. Undoubtedly the plastic deformation observed is due to the distorted shear stress field in the vicinity of a furrow. The deformation takes place progressively with cycling, and in this process the shear stress field is altered in such a way that the maximum shear stress decreases and moves further down beneath the surface. The furrow profile finally stabilizes, when the stress has been reduced to the yield strength level.

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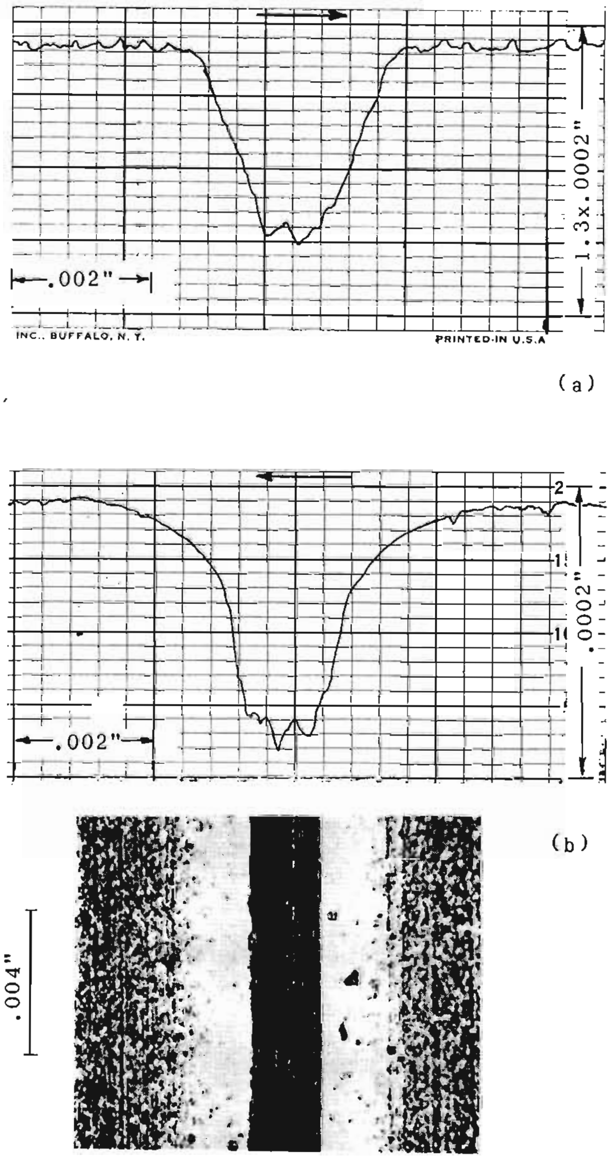

Fig. A2

FIG. A2. (a) Talysurf trace across a surface furrow on an unrun 6309 inner ring. (b) Talysurf trace and microphotograph at the same location after 12.5 millions endurance testing. Material: Bethlehem 52100 CVD bar.

AUTHORS' CLOSURE

The authors concur with the discussor's remarks' and have no word to add to his discussion and analysis.